OO-T-1719A
July 31, 1975
SUPERSEDING
Fed. Spec. OO-T-1719
January 5, 1972

FEDERAL SPECIFICATION

TYING MACHINES, BUNDLE, INDUSTRIAL

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers tying machines for tying bundles.

1.2 Classification.

1.2.1 Types. Tying machines shall be of the following types as specified (see 6.2):

Type I - Single Wrap
Type IA - Single Wrap (Heavy)
Type II - Double Wrap

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

CC-M-636 - Motor, Alternating-Current, (Fractional Horsepower).
CC-M-641 - Motor, Alternating-Current, (Integral Horsepower), (200 HP and Smaller).

Federal Standard:

FED-STD-123 - Marking for Domestic Shipment (Civil agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

FSC 3540

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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-T-152	-	Treatment, Moisture and Fungus Resistant of Communications, Electronic, and Associated Electrical Equipment
MIL-P-514	-	Plates, Identification, Instructions and Marking, Blank
MIL-L-3153	-	Laundry and Dry Cleaning Machinery and Equipment (For Fixed Installations), Preparation for Delivery Of

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-461	-	Electromagnetic Interference Characteristics, Requirements for
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-1474	-	Noise Limits for Army Material

(Copies of Military specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

29 CFR 1910 - OCCUPATIONAL SAFETY AND HEALTH ACT (OSHA) STANDARDS

(The Code of Federal Regulations (CFR) and Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Electrical Manufacturers' Association (NEMA)

ICS-1970 - Industrial Controls and Systems

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 155 East 44th Street, New York, NY 10017.)

Underwriters' Laboratories, Inc. (UL) Standards

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611; 1285 Walt Whitman Road, Melville, L.I., NY 11749; or 1655 Scott Boulevard, Santa Clara, CA 95050.)

National Fire Protection Association (NFPA)

No. 70 - The National Electrical Code (1968)

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, MA 02110.)

American Society for Testing and Materials (ASTM)

A 167 - Corrosion-Resisting Chromium-Nickel Steel Plate, Sheet and Strip
A 176 - Corrosion-Resisting Chromium Steel, Plate, Sheet and Strip

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

American National Standards Institute (ANSI) Inc.

ANSI S1.4 - 1971
Specifications For Sound Level Meters

(Application for copies should be addressed to the American National Standards Institute Inc., 1430 Broadway, New York, NY 10018.)

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3. REQUIREMENTS

3.1 Standard product. The tying machine furnished under this specification shall be the manufacturer's current standard product except for changes necessary to comply with the requirements of this specification.

3.2 Codes and standards.

3.2.1 NFPA. The tying machine shall be wired to conform to NFPA Standard No. 70.

3.2.2 UL. The wiring and conduit shall conform to the applicable Underwriters' Laboratories (UL) Standards for use in nonhazardous locations.

3.2.3 NEMA. Motor controllers and switches shall conform to performance requirements of NEMA Standards Publication No. ICS-1970.

3.2.4 OSHA. The tying machine shall comply with applicable requirements of OSHA Standards 29 CFR 1910 promulgated under the Occupational Safety and Health Act.

3.2.5 Certification. Prior to approval of the first shipment, the supplier shall submit satisfactory evidence to the contracting officer or his authorized representative, that the tying machines he proposes to furnish meet the requirements specified in 3.2.1, 3.2.2, 3.2.3 and 3.2.4.

3.2.5.1 OSHA. Acceptable evidence of meeting the applicable safety and health standards of OSHA shall be a certificate of compliance certifying that the tying machine furnished conforms to OSHA Standards 29 CFR 1910.

3.2.5.2 NFPA. Acceptable evidence of meeting requirements of 3.2.1 shall be the manufacturer's certified statement that the tying machines are wired to conform to NFPA Standard No. 70.

3.2.5.3 UL. Acceptable evidence of meeting requirements of 3.2.2 shall be the UL label or Listing Mark or certification from an independent testing laboratory acceptable to the contracting officer or his authorized representative that the wiring and conduit conform to applicable UL Standards.

3.2.5.4 NEMA. Acceptable evidence of meeting the requirements of 3.2.3 shall be the manufacturer's certified statement that the motor controllers and switches conform to NEMA Standards Publication No. ICS-1970. A tag or label attached to these components stating the component conforms to this standard is acceptable evidence.

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3.3 Materials. Materials not definitely specified shall be of the quality normally used by the manufacturer for tying machines provided the completed items comply with all provisions of this specification.

3.3.1 Corrosion-resisting steel. Corrosion-resisting steel used in the fabrication of the tying machine shall conform to the requirements of ASTM Specifications A 167 or A 176, as applicable.

3.3.2 Conduit and wire electric. Electrical conduit shall be rigid or flexible steel or electrical metallic tubing. All electrical wire shall be heat-resistant grade, thermoplastic insulated.

3.3.3 Motor. The motor shall be supplied with an enclosure in accordance with the manufacturer's standard practice. Motor shall conform to the requirements of CC-M-636 or CC-M-641, as applicable, and shall conform to the following requirements:

(a) The motor shall have windings impregnated to resist moisture.

(b) Motor shall be equipped with ball bearings, except that a motor of 1/2 horsepower and less when used in horizontal applications may have sleeve bearings.

(c) Motor bearings shall be of the permanently lubricated type or shall have adequate and accessible means for lubrication.

(d) A motor of 1/8 horsepower rating or larger shall be provided with thermal-overload protection incorporated as a part of the motor or installed in the controller.

3.4 Design and construction. The industrial bundle tying machines shall be types I, IA or type II, as specified.

(a) Types I and IA machines shall be designed to make one wrap of twine around the bundle and tie and cut twine per cycle of operation.

(b) Type II machine shall be designed to make two wraps of twine around the bundle and tie and cut twine per cycle of operation. The type II machine shall permit the operator to turn the bundle 90 degrees during the wrapping cycle allowing the bundle to be wrapped in two directions prior to tying the knot.

Type IA machine shall be capable of tying bundles with either sisalana or polypropylene twines having knot breaking strengths up to 160 pounds. Types I and II machines shall be capable of tying bundles with a 16-ply cotton wrapping twine or equivalent synthetic material (i.e., polyester twines) having a breaking strength of 30 pounds or more. All knots shall be of nonslip bow type

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and the twine ends shall be cut no less than a minimum of 3/8-inch or a maximum of 1 inch from the knot. The machines shall be self-adjusting to any size of bundle from 4 to 20 inches in height by 8 to 23 inches in width. The assembly item shall show that the assebmly of components has complete and proper inter-relationship, no overheating of parts, starting and stopping devices operate smoothly and positively, and there are no structural failures (see 4.3.1). When tested as specified in 4.3.3, the tying machine noise level shall not exceed category D of MIL-STD-1474.

3.4.1 Control. The control for the wrapping, tying, and cutting of the twine shall be by means of a clutch controlled foot treadle. Unless otherwise specified (see 6.2) the machines shall be provided with an electrical interlock for safe operation of the needle or tying arm. The machines shall be wired in such a manner that the proper sequence must be maintained to operate the needle or tying arm. If the foot treadle is depressed before the power switch is moved to the "Start" or "On" position, the needle or tying arm shall remain at rest position. If the power switch is moved to the "Start" or "On" position and then the foot treadle is depressed, the needle or tying arm shall begin the tying cycle.

3.4.2 Power. The machine shall be electrically powered with an arrangement for continuous operation of the motor. Motor shall conform to the requirements of 3.3.3.

3.4.3 Frame. The frame of the machine shall be ferrous metal and shall be mounted on rubber or composition wheel swivel casters. The frame shall support the electric motor, twine holder, and a corrosion-resisting steel bundle support table.

3.4.4 Safety devices. All exposed belts, shaft, pulleys and other moving parts not within the periphery of the machine frame shall be guarded. Guards shall be cast iron, sheet metal, wire mesh or tempered hardboard, and shall be securely attached and readily removable.

3.4.4.1 Safety guard. The tying machine's needle or tying arm shall be either of an oscillating or rotating design and shall be mechanically guarded to prevent contact or injury to any person by the needle or arm during machine operation. The guard may be steel or high impact plastic. This device is in addition to the safety requirements specified in 3.4.4. The capacity of the tying machine shall not be impaired by this device.

3.4.5 Belts and pulleys. Driving pulleys shall be grooved for the type belt used and pulleys that are being driven may be grooved or not grooved in accordance with commercial practice. Means shall be provided for adjusting belt tension.

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3.4.6 Lubrication. Means shall be provided to insure lubrication of all moving parts without allowing the lubricant to enter the area of bundle tying.

3.4.7 Name and data plates. Tying machines for military procurement shall be marked for identification in accordance with MIL-STD-130 on a plate conforming to MIL-P-514. Tying machines for Civil agency procurement shall be marked in accordance with the manufacturer's standard practice.

3.4.8 Finish. Unless otherwise specified (see 6.2), the tying machine except pushbuttons shall be finished according to the manufacturer's standard commercial practice. Emergency stop and normal stop pushbuttons shall be red and this color shall be limited to pushbuttons or lighted tippette switch.

3.4.9 Electromagnetic compatibility. When specified (see 6.2), equipment shall be designed and equipped for electromagnetic compatibility in accordance with class IIB of MIL-STD-461 (see 4.3.2).

3.4.10 Moisture and fungus resistant treatment. When specified (see 6.2), electrical components of tying machines covered by this specification shall be moisture and fungus resistant treated with material conforming to and applied as specified in MIL-T-152.

3.4.11 Electrical characteristics. The electrical parts of the machine shall be suitable for the voltage, frequency, and number of phases for alternating current (AC) or voltage for direct current (DC) as specified (see 6.2).

3.5 Workmanship. The completely fabricated tying machine shall be free from defects such as fractures, splits, punctures, tears, dents, creases, bows, miscasts, deteriorations, or malformations. There shall be no sharp edges, burrs, or slivers.

3.5.1 Application of finish. The finish applied to the end item shall be free from blisters, peeling or areas of no film and shall not be chipped.

3.5.2 Threaded fasteners. Threaded fasteners shall not be broken, stripped, fractured, or loose.

3.5.3 Rivets. Rivets shall be driven to completely fill the holes. Rivet heads shall be in full contact with the surface of the riveted members and shall not be loose.

3.5.4 Electrical wiring. Wiring shall not be cut, abraded, or have excess insulation stripped, and shall be securely joined at terminals. Wiring shall have slack to provide strain relief.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification, or applicable purchase document.

4.2.2 End item inspection. A lot shall consist of all tying machines of one type offered for inspection at one time. The sample unit for this inspection shall be one completely fabricated tying machine.

4.2.2.1 Visual examination. Examination shall be made of the end item for defects listed in table II. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Types I and II	Finish not color and type as specified		X
	Blistered, peeled, chipped, or area of no film		X
Construction and workmanship	Fractured, split, punctured, dented, or deteriorated	X	
	Sharp burr, edge, sliver, or splinter	X	
	Component loose		X

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TABLE I. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Construction and workmanship (cont'd)	Not connected or joined as specified	X	
	Any component not readily accessible for servicing where required		X
	Moving parts not guarded	X	
	Design not for making one or two wraps across or lengthwise around the bundle per tie and cut as specified	X	
	Knots not made of nonslip bow type and twine ends longer or shorter than specified		X
	Control for tying and cutting of twine not by foot treadle	X	
	Not electrically powered with arrangements for continuous operation of the motor	X	
	Not type specified	X	
	Not wired for sequence operation	X	
	Not properly guarded	X	
Thread fasteners	Missing, broken, stripped, fractured	X	
	Loose		X
Rivets (when applicable)	Broken, loose, not peened, or insufficiently peened	X	
Electrical assembly	Wiring cut, abraded, not properly joined, loose at terminal or not inclosed in specified conduit where required	X	
	Adequate slack not provided for wiring to relieve strain or insulation stripped from wiring	X	
Lubrication (where applicable)	Not lubricated where required	X	
	Lubrication fitting not accessible for servicing		X
Marking for identification	Missing, incomplete, not legible		X

4.2.2.2 Testing of the end item. Each unit of production shall be tested as specified in 4.3.1. Any nonconformance shall be cause for rejection of

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the unit. The initial unit of production shall be tested as specified in 4.3.1 and 4.3.3 and when applicable, 4.3.2. Any nonconformance shall be cause for rejection of the lot.

4.2.2.3 Code and standards compliance. When moisture and fungus resistant treatment is specified, the supplier shall furnish a certificate of compliance that electrical components comply with 3.4.10. Proof of compliance with the requirements of 3.2 shall be made available to the Government representative.

4.2.3 Examination of preparation for delivery. Examination of preparation for delivery shall be in accordance with MIL-L-3153.

4.3 Tests.

4.3.1 Operational testing. Each type I or type II tying machine as applicable shall be tested by the supplier by operating with representative bundles and with twine for a period of 15 minutes to determine that the wrapping mechanisms functions satisfactorily. Any nonconformance with requirements of 3.4 shall constitute failure of the test.

4.3.2 Electromagnetic compatibility. When electromagnetic compatibility is required, the initial unit of production shall be tested by the supplier in accordance with test methods CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the supplier or an independent testing agency. The supplier shall furnish the contracting office written certification that the Interference Control Plan, The EMI/EMC Test Plan, the Electromagnetic Test report and the requirements meet MIL-STD-461

4.3.3 Noise evaluation test. The noise evaluation test shall be conducted as follows:

4.3.3.1 Instrumentation. A sound level meter meeting the requirements of ANSI S1.4 - 1971 Specifications for Sound Level Meters shall be used for survey measurements. An acoustical calibrator shall be used for verifying amplified gain and absolute sound level of the sound level meter prior to and following each series of measurements.

4.3.3.2 Acoustical environment. The test site shall consist of an outdoor open space area of uniform grade and free of reflecting surfaces (e.g., buildings, trees, or hill sites) within a minimum distance of 50 feet (15 meters) from the equipment under test. The ground surface between the microphone and the equipment being tested shall be preferably concrete, asphalt, or similar hard material free of snow, long grass, or loose soil. Bystanders in the vicinity of either the microphone or test object may appreciably influence the meter reading, therefore, only the person reading the meter, an optional observer, and the person operating the equipment being tested should be within the test site. The observer, if used, shall be directly behind the person

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reading the meter. The level of ambient noise, including wind effects, must be 10dBA lower than that produced by the equipment being tested. The observer should insure that wind gusts do not influence the measurements, and no measurements should be made when wind velocity is in excess of 12 miles per hour.

4.3.3.3 Measurement. The microphone shall be located at the operator's position, 4 feet above the ground plane with the direction of the microphone oriented as specified in the meter manufacturer's operating manual.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A and C. Each tying machine shall be preserved and packaged in accordance with the applicable requirements of MIL-L-3153.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C. Tying machines, preserved and packaged as specified in 5.1, shall be packed in accordance with the applicable requirements of MIL-L-3153.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with the applicable requirements of FED-STD-123.

5.3.2 Military requirements. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with the applicable requirments of MIL-L-3153.

6. NOTES

6.1 Intended use. The industrial bundle tying machines covered by this specification are intended for use in fixed laundry installations.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2.1).
(c) When interlock is not required (see 3.4.1).
(d) Finish, if other than manufacturer's standard is required (see 3.4.8).
(e) When electromagnetic compatibility is required (see 3.4.9).

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(f) When moisture and fungus-resistant treatment is required (see 3.4.10).
(g) Whether alternating current or direct current is required (see 3.4.11).
(h) Selection of applicable levels of preservation, packaging, and packing (see 5.1 and 5.2).

6.2.1 Contract data requirements. Any data items to be delivered under any contract for items covered by this specification should be specifically called for in the contract in accordance with the applicable regulations of the procuring activity.

6.3 New type IA was added for this revision.

Custodians:

Army - GL
Navy - MC
Air Force - 84

Review activity:

Army - MD

User activity:

Army - SM

Preparing activity:

Army - GL

Civil Agency Coordinating Activities:

GSA-FSS
GPO

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